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1.Project Overview

This project analyzes telecom customer data to understand behavior patterns and predict churn. Phase 1 focuses on data cleaning, exploratory data analysis (EDA), and extracting business insights that will inform a predictive model in Phase 2.

2.Problem Statement

Customer churn significantly impacts business revenue and customer retention. Organizations require predictive analytics solutions that can identify customers at risk of leaving so that targeted retention strategies can be implemented.

3.Objectives

- Analyze customer demographic and service data.
- Clean and preprocess the dataset.
- Perform exploratory data analysis.
- Identify factors influencing customer churn.
- Build predictive machine learning models.
- Evaluate model performance.
- Provide business recommendations.

4.Dataset Description

4.1 Dataset Structure

The dataset contains 7,043 customer records with 33 original features, including demographics (gender, senior citizen status, partner/dependents), account details (tenure, contract, payment method), service subscriptions (internet, phone, streaming, security add-ons), billing information (monthly/total charges), and churn outcome fields (churn label, score, reason, and CLTV).

- Records: 7,043 unique customers (no duplicate Customer IDs found)
- Feature types: categorical (services, contract, payment), numeric (tenure, charges, CLTV, churn score), geographic (city, state, lat/long)
- Target variable: Churn Label / Churn Value (binary)

5.Tools and Technologies

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Excel
- GitHub

6.Methodology

The methodology adopted for this project follows a structured data analytics and machine learning workflow. The process begins with collecting customer data, followed by data preprocessing, exploratory data analysis, feature engineering, predictive model development, model evaluation, and finally generating business recommendations. Each stage is designed to transform raw customer data into meaningful insights that support customer retention strategies.

Workflow Diagram

Customer Dataset

↓

Data Collection

↓

Data Cleaning & Preprocessing

↓

Exploratory Data Analysis (EDA)

↓

Feature Engineering

↓

Machine Learning Model Building

↓

Model Evaluation

↓

Business Insights & Recommendations

6.1 Data Collection

The first stage of the project involved collecting the Telco Customer Churn dataset. The dataset contains customer demographic information, subscribed services, billing details, account information, and churn status. It consists of 7,043 customer records with 33 attributes, making it suitable for customer behavior analysis and churn prediction. The dataset includes numerical, categorical, and geographical variables, providing a comprehensive view of customer characteristics.

Activities Performed

- Obtained the Telco Customer Churn dataset.
- Loaded the dataset into Python using the Pandas library.
- Inspected the dataset structure.
- Examined the number of rows and columns.
- Identified feature names and data types.
- Verified the target variable (Churn Label).

Purpose

The objective of this stage was to gather reliable customer information for further analysis and predictive modeling.

6.2 Data Cleaning and Preprocessing

Raw datasets often contain missing values, duplicate records, inconsistent formats, and unnecessary attributes that can negatively impact model performance. Therefore, data preprocessing was performed to improve data quality before analysis.

The following preprocessing tasks were carried out:

➢ Missing Value Treatment

The Total Charges column contained blank values for newly joined customers with zero tenure. These values were converted into numeric format and replaced with appropriate values to maintain consistency.

➢ Handling Null Values

The Churn Reason column contained missing values for active customers. Since these customers had not left the company, the missing values were replaced with "Not Applicable" instead of leaving them blank.

➢ Duplicate Record Checking

The Customer ID column was checked for duplicate records. No duplicate customer records were found in the dataset.

➢ Removing Unnecessary Features

Redundant columns such as Count and Lat Long were removed because they provided no additional analytical value.

➢ Data Type Conversion

Several columns were converted into appropriate numerical and categorical formats to facilitate statistical analysis and machine learning.

Purpose

The purpose of preprocessing was to ensure that the dataset was clean, consistent, and suitable for exploratory analysis and predictive modeling.

6.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand customer behavior, identify patterns, detect trends, and discover relationships between different variables.

Various statistical summaries and visualizations were created using Matplotlib and Seaborn.

The analysis included:

- Customer churn distribution
- Customer tenure analysis
- Monthly charges distribution
- Contract type analysis
- Internet service analysis
- Payment method analysis
- Customer Lifetime Value (CLTV)
- Correlation analysis
- Demographic analysis
- Add-on services analysis
- Customer segmentation

Visualizations such as bar charts, count plots, histograms, box plots, and heatmaps were generated to understand customer behavior. The EDA revealed important churn drivers including contract type, tenure, monthly charges, payment methods, and service subscriptions.

Purpose

The main objective of EDA was to identify hidden patterns and determine which customer attributes have the greatest influence on churn.

6.4 Feature Engineering

Feature engineering improves the predictive capability of machine learning models by creating meaningful input variables.

The following activities were performed:

- Converted categorical variables into numerical values using encoding techniques.
- Created customer value segments based on Customer Lifetime Value (CLTV).

- Selected the most relevant features affecting customer churn.
- Removed irrelevant attributes that did not contribute to prediction.
- Prepared the feature matrix (X) and target variable (y).

Purpose

Feature engineering enhances model accuracy by providing informative and relevant features while reducing unnecessary information.

6.5 Model Building

After preprocessing and feature engineering, machine learning models were developed to predict customer churn.

The dataset was divided into training and testing subsets using an 80:20 ratio.

Several classification algorithms were considered, including:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- Gradient Boosting Classifier (optional)

The models were trained using historical customer data to classify whether a customer is likely to churn or remain with the company.

Purpose

The objective of model building was to create an accurate predictive model capable of identifying customers who are at risk of leaving.

6.6 Model Evaluation

The performance of the trained models was evaluated using standard classification metrics.

The evaluation included:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score
- Confusion Matrix

These metrics were used to compare different machine learning models and identify the best-performing model for churn prediction.

Purpose

The goal of model evaluation was to measure prediction performance and ensure that the selected model generalizes well to unseen customer data.

6.7 Business Recommendations

Based on the exploratory analysis and predictive modeling results, several business recommendations were proposed to reduce customer churn and improve customer retention.

Key recommendations include:

- Focus retention efforts on new customers during their first two years.
- Encourage customers to switch from month-to-month contracts to long-term plans.
- Improve the quality of customer support services.
- Promote automatic payment methods to reduce churn.
- Bundle value-added services such as online security and technical support.
- Prioritize retention campaigns for high-value customers with high churn risk.
- Improve customer satisfaction by addressing service quality and pricing concerns.

Purpose

The recommendations help organizations make data-driven decisions that improve customer satisfaction, increase loyalty, and reduce revenue loss caused by customer churn.

5.Data Cleaning and Preprocessing

Data Cleaning

The following data quality issues were identified and resolved:

- 'Total Charges' contained 11 blank/whitespace entries for customers with zero tenure (new sign-ups); these were converted to numeric and filled with 0 or the equivalent monthly charge.
- 'Churn Reason' was null for all active (non-churned) customers by design; these were explicitly labeled 'Not Applicable' rather than left as missing.
- Redundant/constant columns ('Count', 'Lat Long' string) were dropped since 'Latitude' and 'Longitude' already exist as numeric fields.
- No duplicate customer records were found in the dataset.

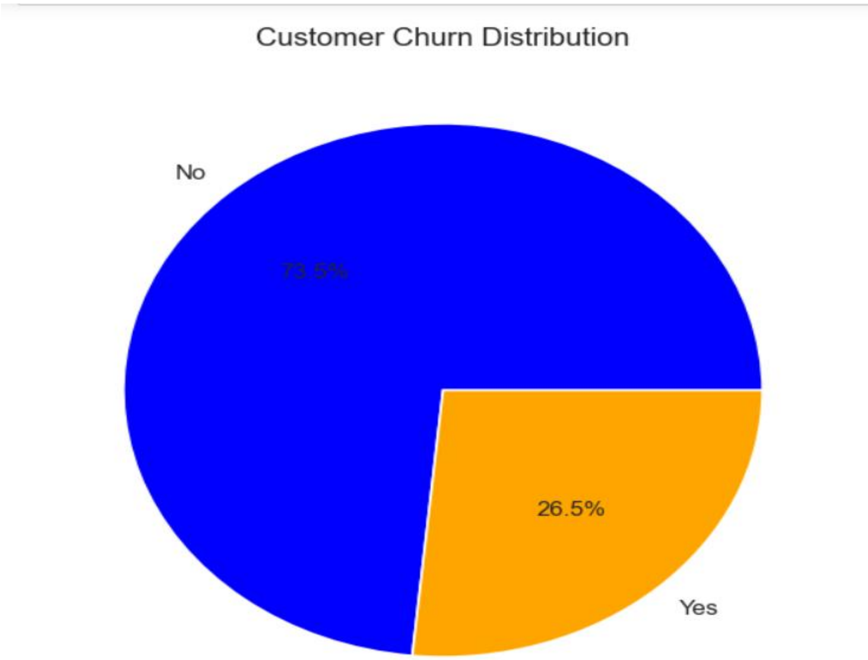
The cleaned dataset (7,043 rows × 31 columns) was exported as telco_churn_cleaned.csv for downstream modeling.

Key Metrics Summary

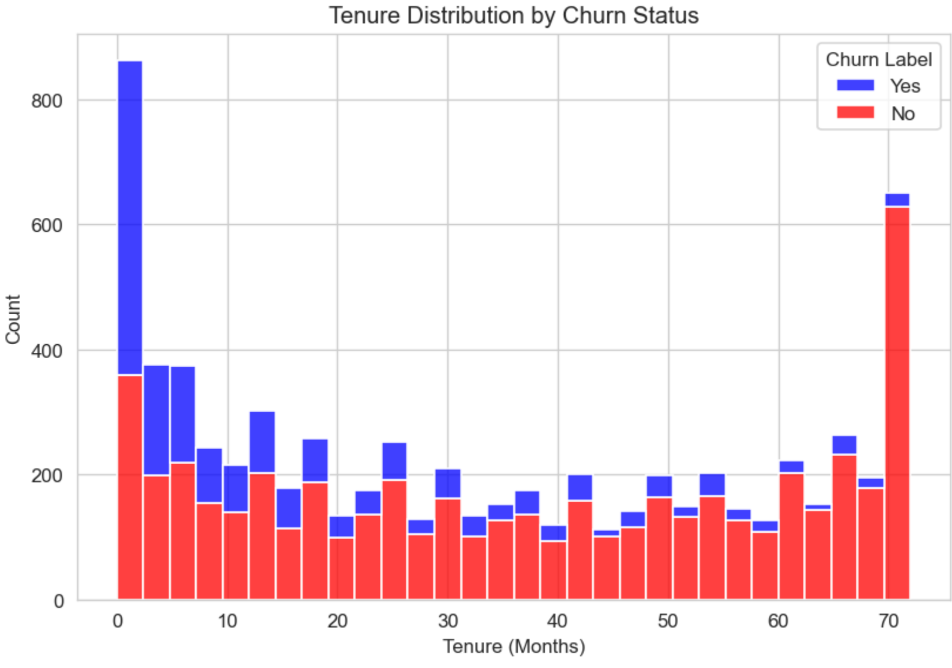
Metric	Value
Overall Churn Rate	26.5%
Avg. Tenure - Churned Customers	18.0 months
Avg. Tenure - Retained Customers	37.6 months
Avg. Monthly Charges - Churned	$74.4
Avg. Monthly Charges - Retained	$61.3
Avg. CLTV - Churned Customers	$4149.4
Avg. CLTV - Retained Customers	$4490.9
Top Reported Churn Reason	Attitude of support person (192 customers)

Exploratory Data Analysis (EDA)

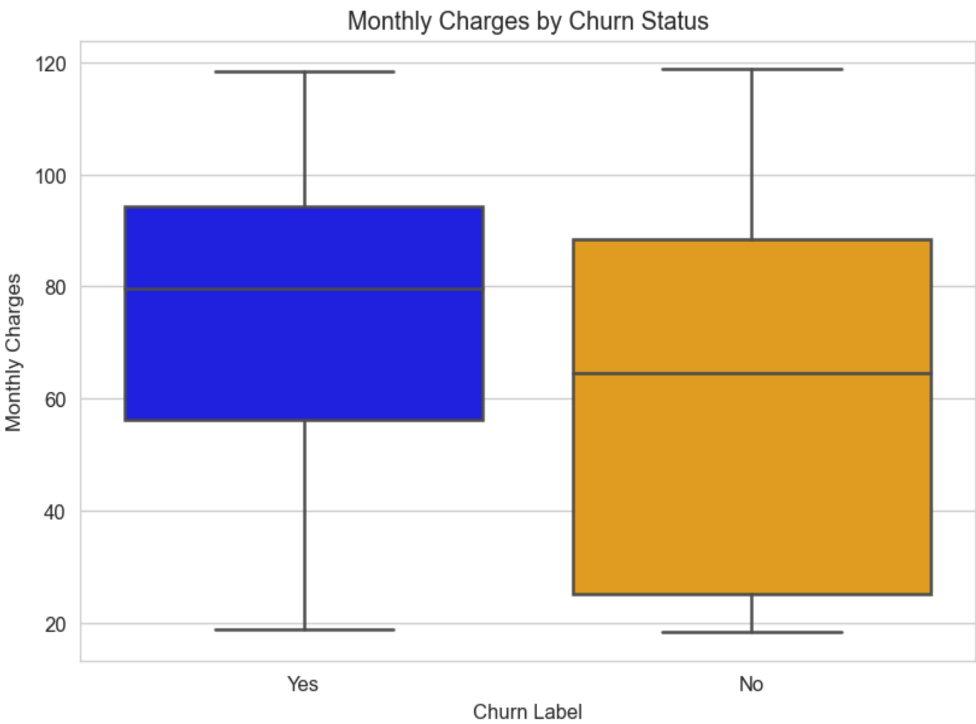
8.1 Customer Churn Distribution



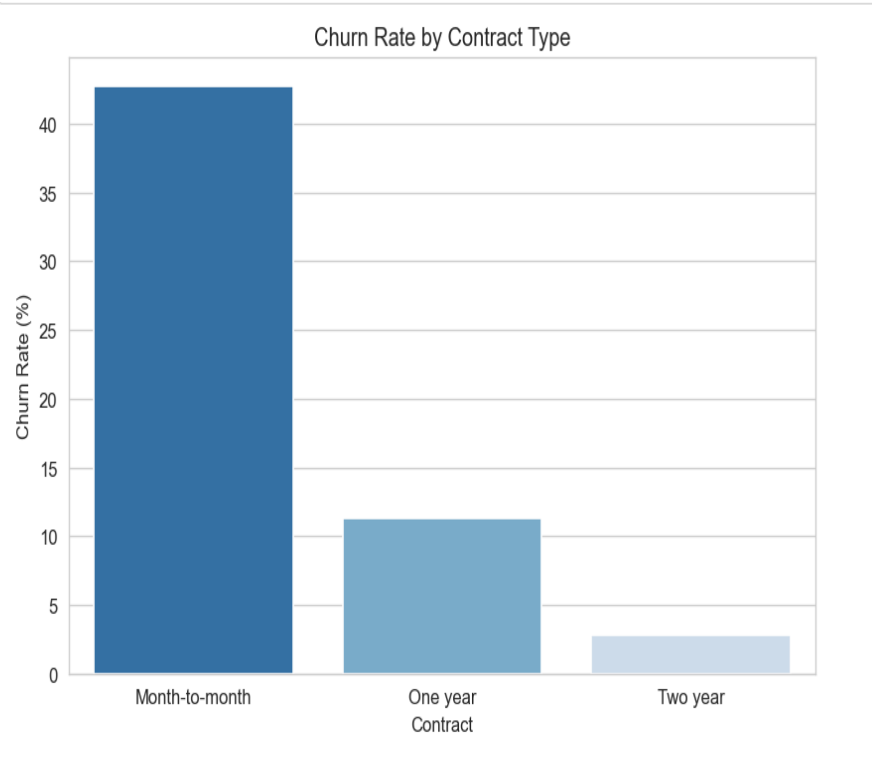
8.2 Customer Tenure Analysis



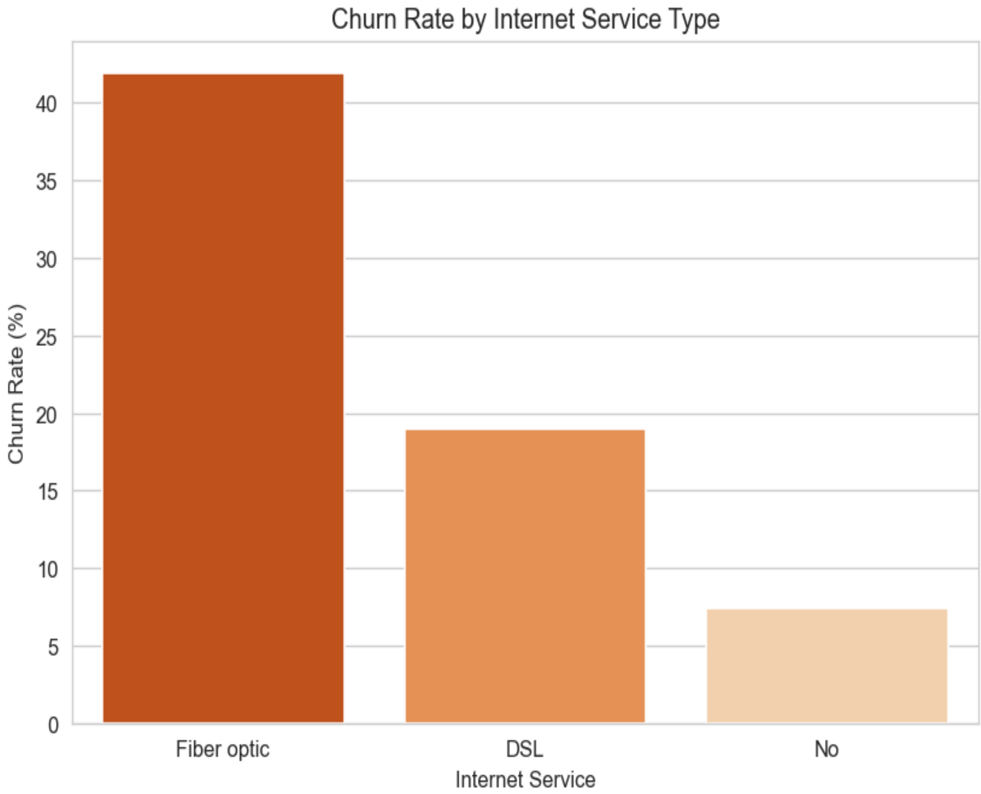
8.3 Monthly Charges Analysis



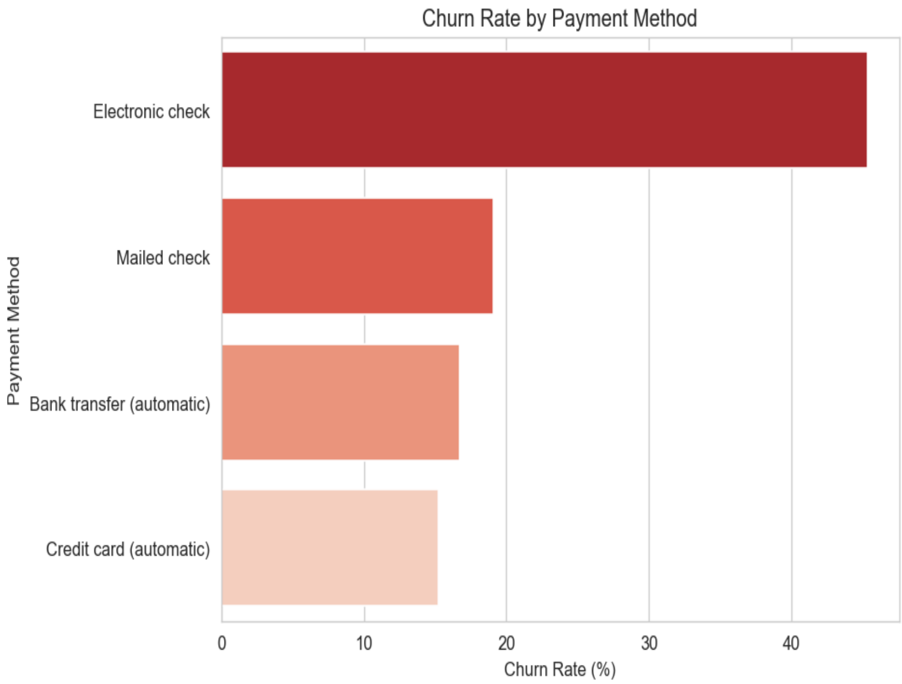
8.4 Contract Type Analysis



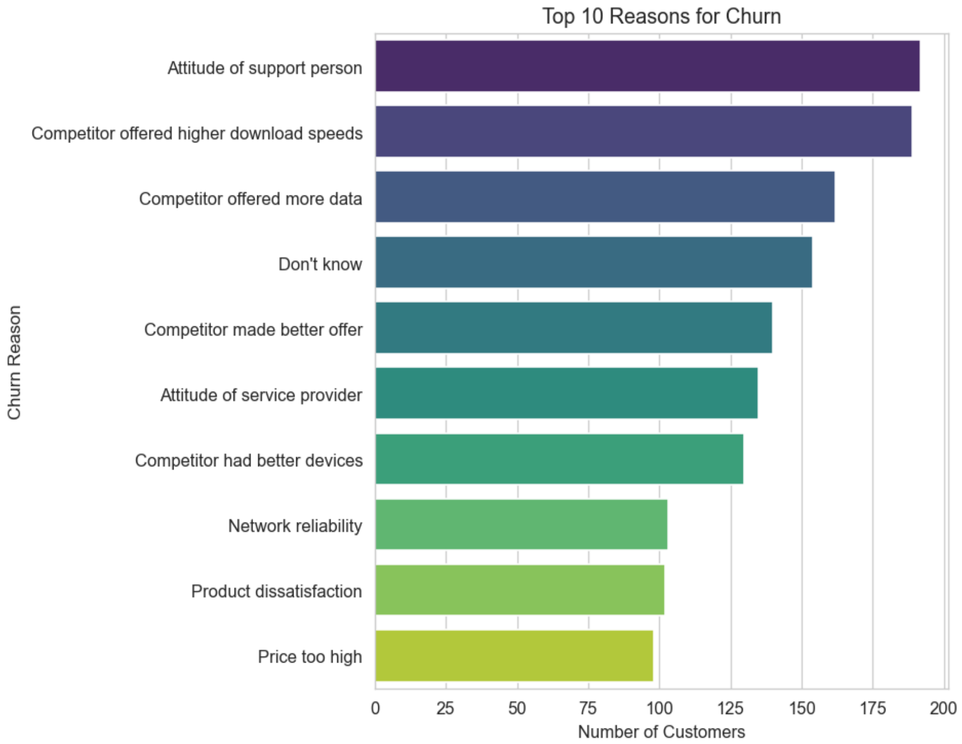
8.5 Internet Service Analysis



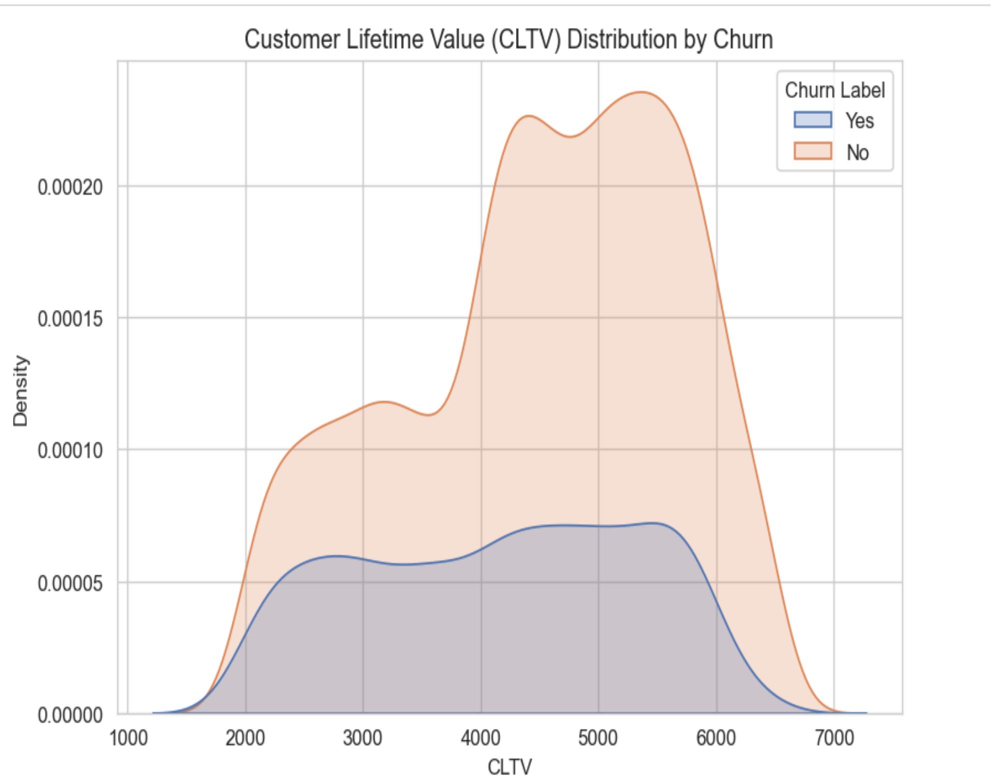
8.6 Payment Method Analysis



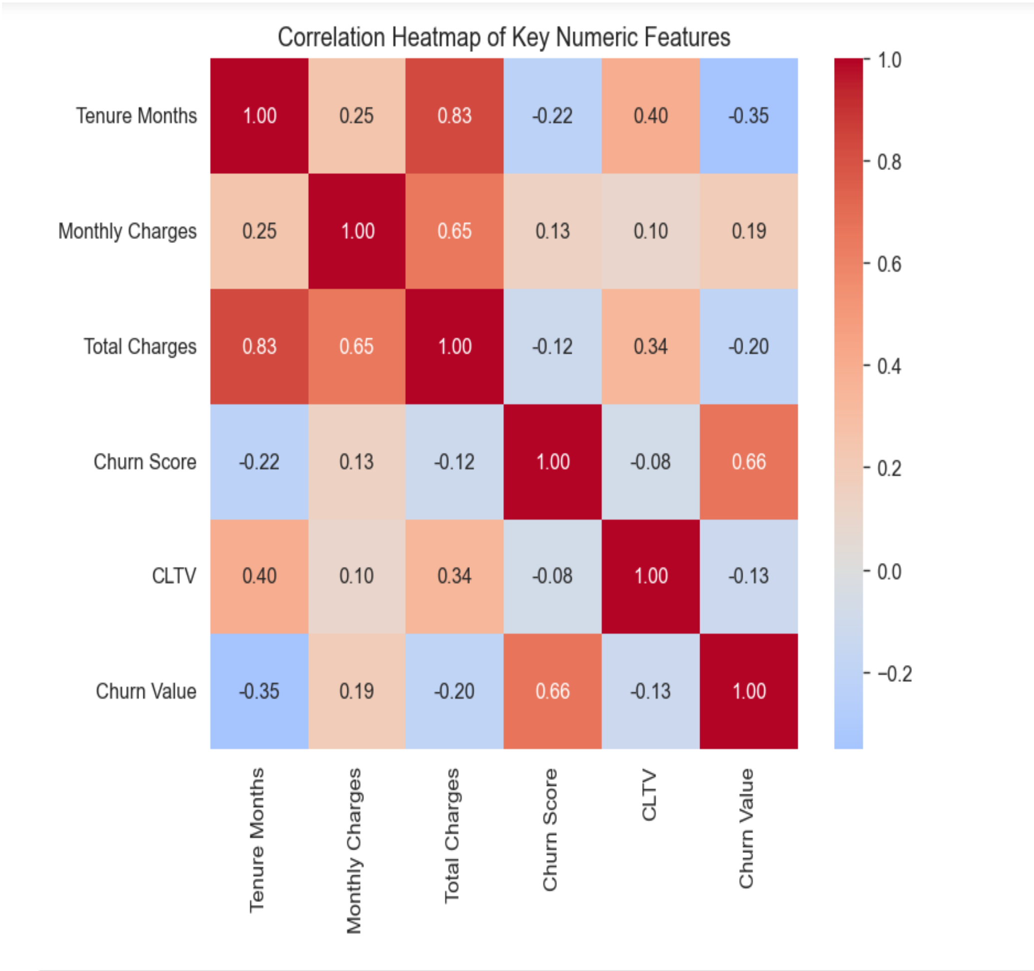
8.7 Churn Reasons Analysis



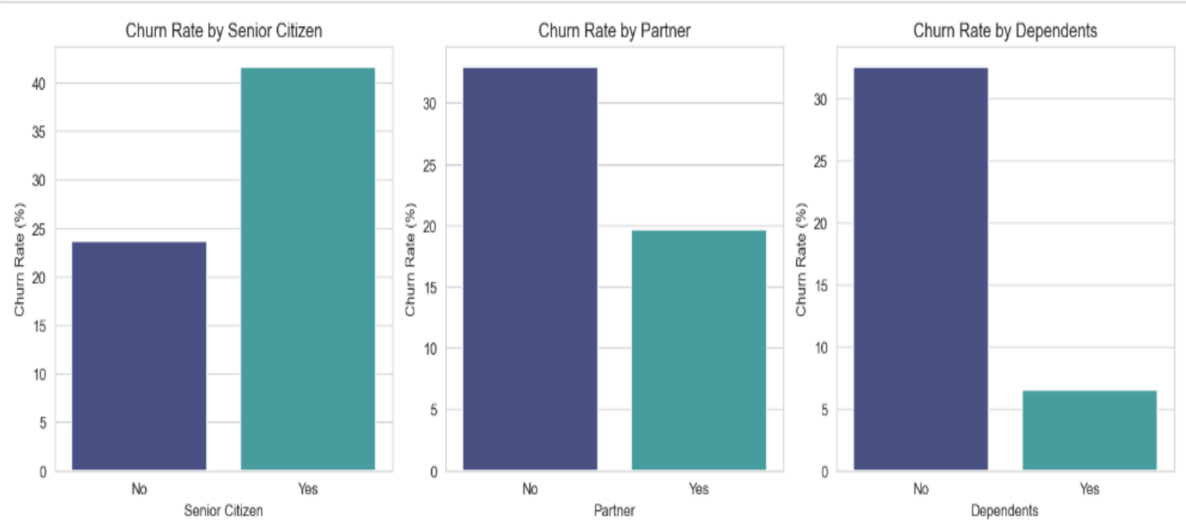
8.8 Customer Lifetime Value Analysis



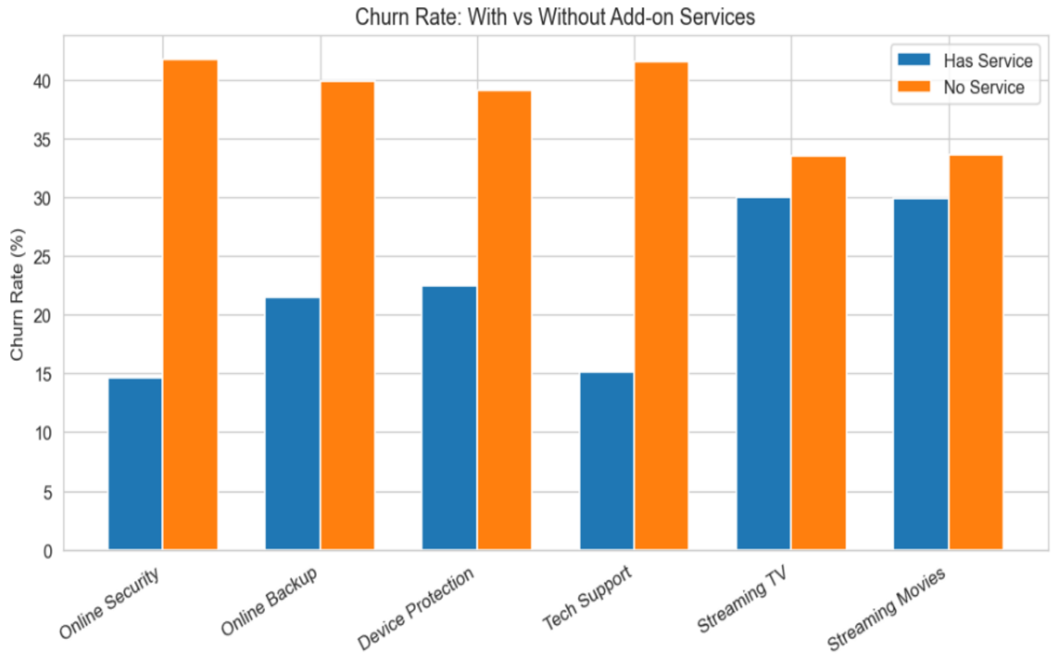
8.9 Correlation Analysis



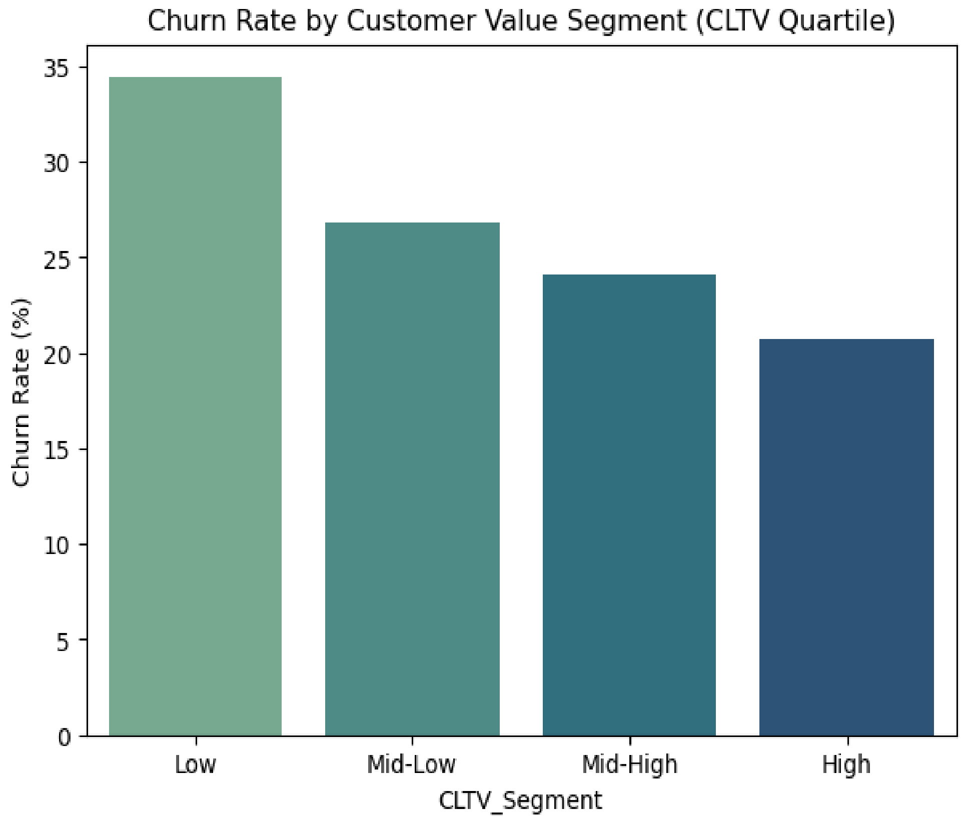
8.10 Demographic Analysis



8.11 Add-on Services Analysis



8.12 Customer Segmentation



9.Key Findings

The EDA revealed a 26.5% churn rate, with churned customers having lower average tenure (18 months) and higher monthly charges ($74.4). Month-to-month contracts, electronic check payments, and fiber optic services showed higher churn. Poor customer support was the leading churn reason, while high-CLTV customers demonstrated greater loyalty.

Phase 2: Predictive Modeling & Customer Insights

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1. Objective

Phase 1 of this project focused on cleaning the IBM Telco Customer Churn dataset and performing exploratory data analysis. Phase 2 builds on this foundation by engineering predictive features, training multiple classification models to predict customer churn, evaluating and comparing their performance, and translating the model results into actionable business recommendations for customer retention.

2. Feature Engineering

Beyond the cleaned Phase 1 dataset, the following derived features were engineered to capture customer loyalty, value, and engagement patterns relevant to churn:

- Tenure Group — customers bucketed into 0-1yr, 1-2yr, 2-4yr, 4-5yr and 5yr+ loyalty bands based on Tenure Months.
- Average Monthly Spend — Total Charges divided by tenure, capturing true spend rate independent of how long a customer has stayed.
- Service Count — number of add-on services subscribed (Online Security, Backup, Device Protection, Tech Support, Streaming TV/Movies), as a proxy for customer engagement/stickiness.
- CLTV Segment — customers split into Low / Medium / High / Premium quartiles of Customer Lifetime Value.
- High Monthly Spend flag — whether a customer's monthly bill exceeds the median, useful for targeting premium retention offers.
- Has Multiple Services flag — whether a customer bundles phone and internet service together.

Two columns present in the raw data — Churn Score and Churn Reason — were deliberately excluded from modeling. Churn Score is IBM's own pre-computed churn propensity score and would leak the answer directly into the model; Churn Reason is only populated after a customer has already churned. Including either would produce an unrealistically inflated (and practically useless) model.

3. Encoding & Scaling

Binary categorical fields (Yes/No, Male/Female, etc.) were label-encoded to 0/1. Multi-category fields (Contract, Internet Service, Payment Method, Multiple Lines, Tenure Group, CLTV Segment, and the individual add-on service columns) were one-hot encoded, expanding the feature set from 18 raw categorical + 8 numeric columns to 42 model-ready features. Numeric features (Tenure Months, Monthly Charges, Total Charges, CLTV, Avg Monthly Spend, Service Count) were standardized using a StandardScaler fit on the training split only, to prevent data leakage into the test set. Data was split 80/20 into training and test sets using stratified sampling to preserve the ~26.5% churn rate in both partitions.

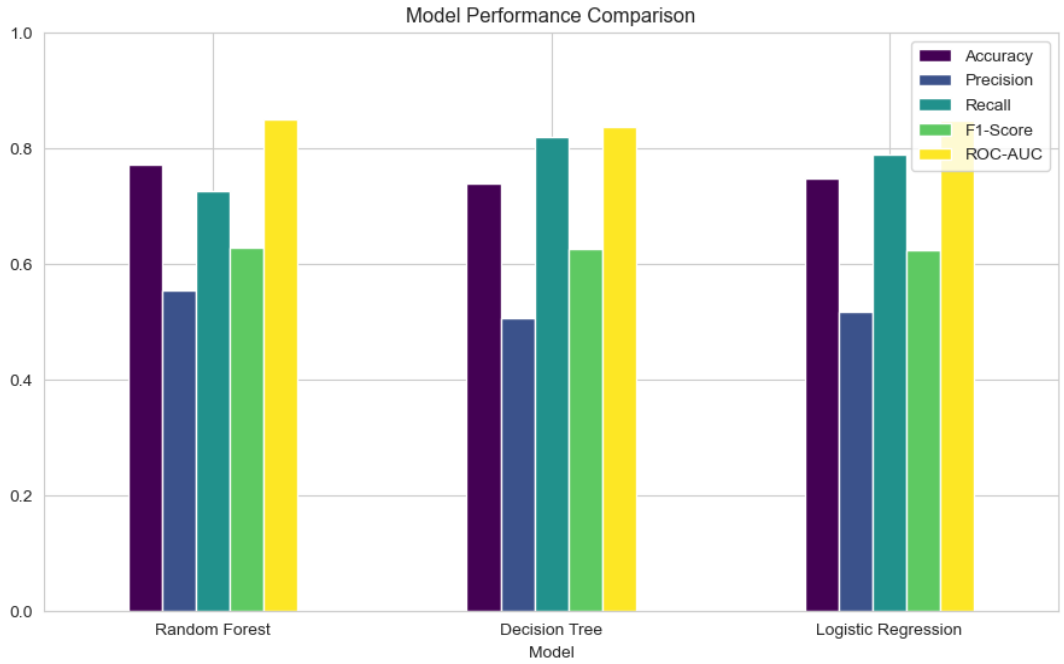
4. Classification Models

Three classification algorithms were trained and compared to predict the binary target Churn Value (1 = churned, 0 = retained). Class weighting was applied to all models to correct for the ~3:1 imbalance between retained and churned customers.

- Logistic Regression — a linear baseline that also yields interpretable, directionally-signed coefficients.
- Decision Tree (max depth 6) — captures non-linear interactions with a human-readable rule structure.
- Random Forest (200 trees, max depth 10) — an ensemble method expected to generalize best and provide robust feature importances.

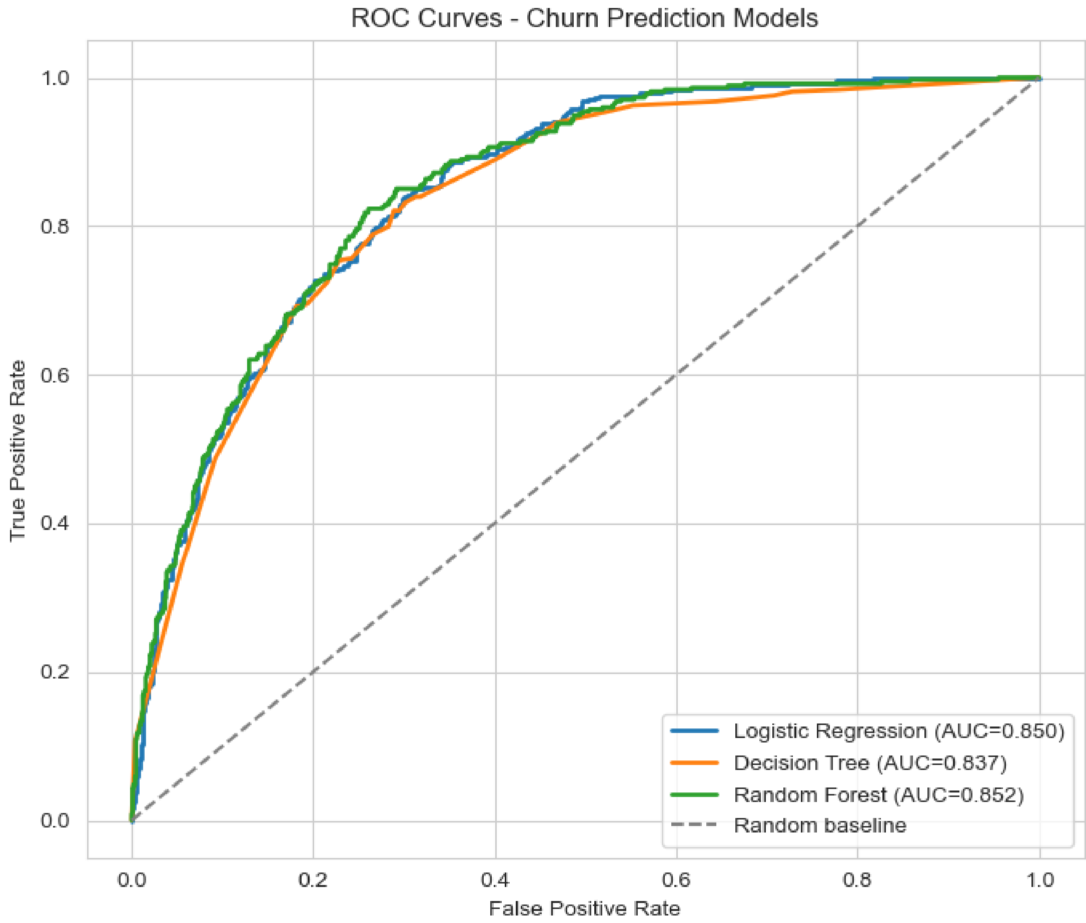
4.1 Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Random Forest	0.773	0.555	0.727	0.630	0.852
Logistic Regression	0.750	0.519	0.791	0.627	0.850
Decision Tree	0.740	0.507	0.821	0.627	0.837



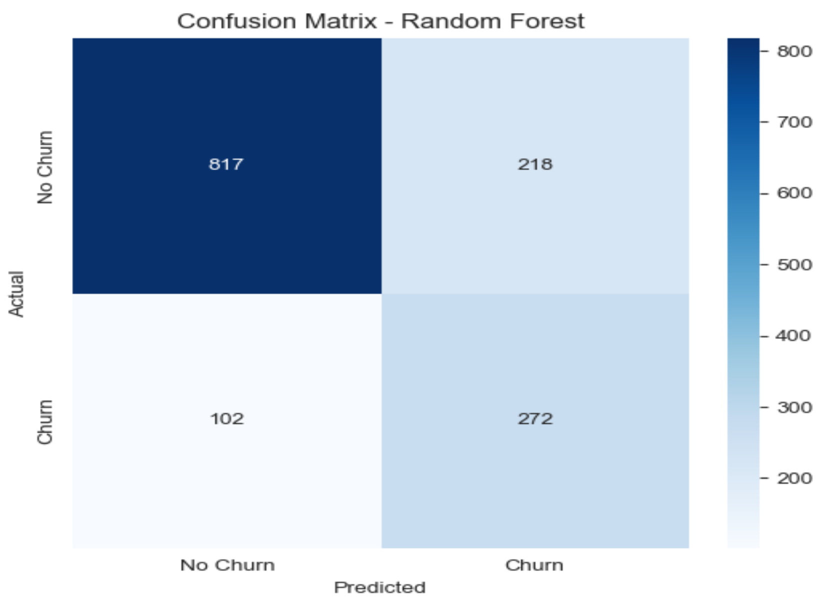
4.2 ROC Curve Analysis

All three models comfortably beat random guessing, with ROC-AUC scores clustered tightly between 0.84 and 0.85 — indicating the churn signal in the engineered features is strong and consistently captured across model types, not an artifact of any single algorithm.



4.3 Confusion Matrix — Best Model (Random Forest)

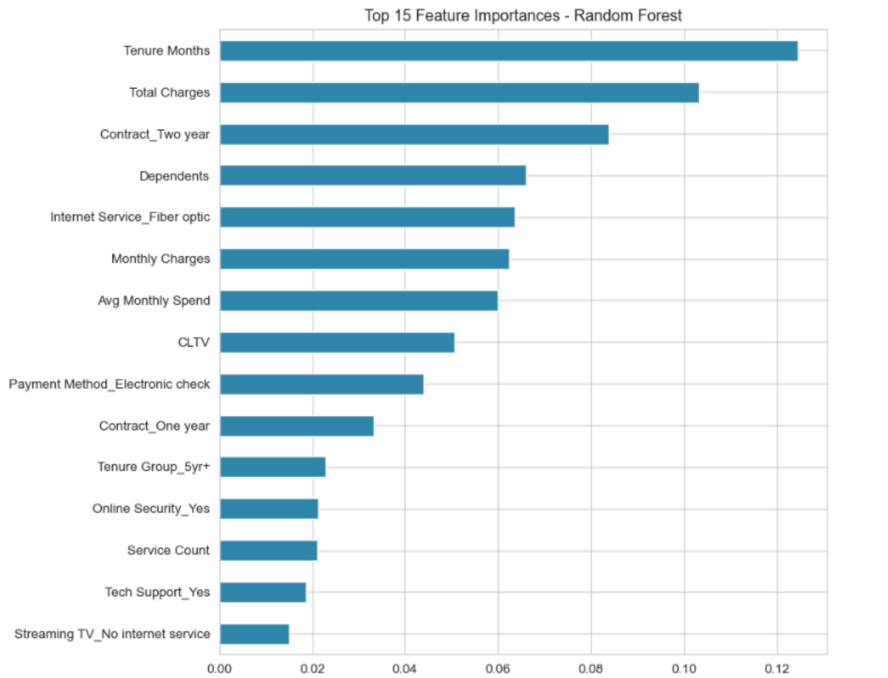
Random Forest correctly identifies 272 of 374 actual churners in the test set (72.7% recall) while keeping false alarms manageable (218 customers flagged as at-risk who did not actually churn). In a retention context, this recall level is valuable because the cost of missing a churner (lost revenue) typically outweighs the cost of a false positive (an unnecessary retention offer).



5. Model Interpretation — Key Drivers of Churn

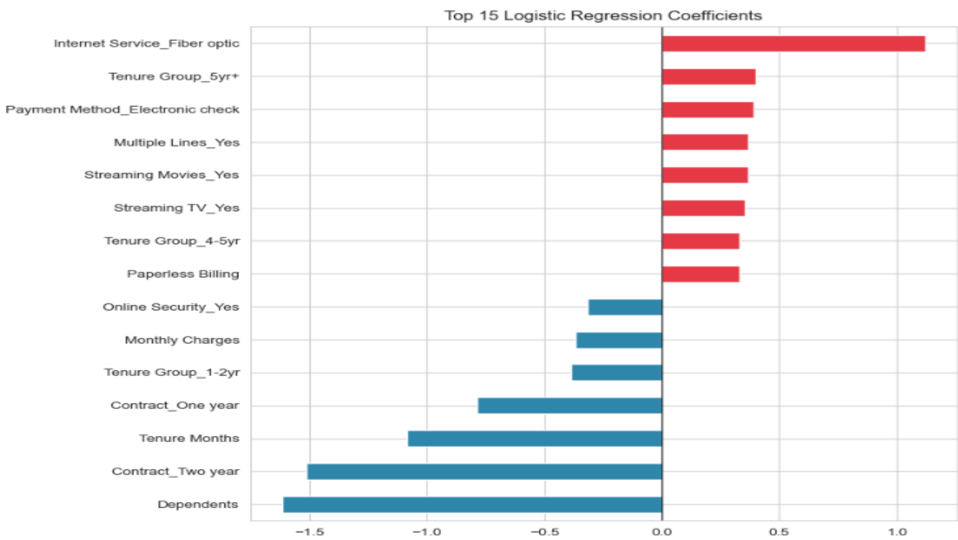
5.1 Random Forest Feature Importance

The Random Forest's built-in feature importance ranks Tenure Months, Total Charges, having a Two-year contract, Dependents, Fiber-optic internet, Monthly Charges, and Average Monthly Spend as the strongest predictors of churn — confirming that how long a customer has stayed and how they're contracted matter more than any single service feature.



5.2 Logistic Regression Coefficients — Direction of Effect

The Logistic Regression coefficients make the direction of each effect explicit. Fiber-optic internet, month-to-month contracts, and Electronic Check payment are associated with a higher churn probability (red), while longer tenure, Two-year contracts, and having Dependents/Partner are associated with lower churn probability (blue).

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6. Segment-Level Churn Insights

Cross-tabulating churn against the engineered segments surfaces clear, actionable risk patterns:

Segment	Category	Churn Rate
Contract Type	Month-to-month	42.7%
Contract Type	One year	11.3%
Contract Type	Two year	2.8%
Tenure Group	0-1 yr (new customers)	47.4%
Tenure Group	5+ yr (loyal customers)	6.6%
Payment Method	Electronic check	45.3%
Payment Method	Credit card (automatic)	15.2%
Internet Service	Fiber optic	41.9%
Internet Service	DSL	19.0%
CLTV Segment	Low CLTV	34.4%
CLTV Segment	Premium CLTV	20.7%

Overall churn rate across the full customer base is 26.5%. Customers on month-to-month contracts paying by electronic check, with fiber-optic internet and under one year of tenure, represent by far the highest-risk segment. Monthly recurring revenue currently at risk from already-churned customers totals approximately $139,131.

7. Business Recommendations

7.1 Customer Retention Strategies

- Prioritize retention outreach to month-to-month, fiber-optic, electronic-check customers in their first 12 months — this segment combines the three strongest churn drivers found in both models.
- Offer contract-upgrade incentives (e.g., a discount for switching month-to-month → one/two-year) since two-year contract customers churn at under 3% versus 42.7% for month-to-month.
- Nudge Electronic Check payers toward automatic bank transfer or credit card billing, which correlates with roughly a 3x lower churn rate — likely because autopay removes a recurring friction point where customers reconsider the service.
- Deploy the Random Forest model as an early-warning system, scoring active customers monthly and routing the top-decile risk scores to a retention team before contract renewal.

7.2 Personalized Marketing Approaches

- For new customers (0-12 months tenure), front-load onboarding support and proactively resolve fiber-optic service issues, since this cohort shows the highest churn rate (47.4%) of any tenure band.

- Bundle add-on services (Online Security, Tech Support) into fiber-optic plans at a discount — the data shows churn drops as Service Count rises, suggesting bundled engagement increases stickiness.
- Segment marketing messaging by CLTV tier: Low-CLTV customers need value-reinforcement messaging, while Premium-CLTV customers warrant loyalty/VIP-style retention perks given their outsized revenue contribution.

7.3 Targeting High-Value Customers

- Even though Premium-CLTV customers churn least (20.7%), they still represent the single largest revenue loss per departure — assign them a dedicated account-health check-in ahead of contract renewal.
- Cross-sell additional services to High/Premium CLTV customers who currently have low Service Count, since higher service adoption is one of the more protective factors against churn.

7.4 Improving Customer Engagement

- Use Service Count and Has Multiple Services as engagement KPIs on internal dashboards — declining engagement scores can serve as an early behavioral signal ahead of an explicit cancellation request.
- Investigate fiber-optic service quality specifically: its churn rate (41.9%) is more than double DSL (19.0%), suggesting a service-quality or pricing issue with the fiber product itself rather than a customer-profile issue.
- Re-run this pipeline periodically (quarterly) with fresh data — churn drivers and segment risk profiles can shift as pricing, competitor offers, and network quality evolve.

8. Conclusion

Random Forest was the best-performing classifier (Accuracy 77.3%, ROC-AUC 0.852), narrowly ahead of Logistic Regression and Decision Tree, though all three models agree on the dominant churn drivers: contract type, tenure, internet service type, and payment method. This convergence across different algorithm families gives confidence that the identified drivers reflect genuine customer behavior patterns rather than model-specific artifacts, and the recommendations above translate directly into a prioritized, data-driven retention roadmap.